KENYA MEDICAL TRAINING COLLEGE
FACULTY OF MEDICAL EDUCATION AND INFORMATION TECHNOLOGY
DEPARTMENT OF HEALTH RECORDS AND INFORMAION

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN HEALTH RECORDS AND INFORMATION TECHNOLOGY

PAPER: HEALTH STATISTICS

DATE: 14th February 2022	**TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

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PART: A: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. What level of measurement would be used if students were asked to choose their favorite color for a set of five?
   A. Ratio
   B. Interval
   C. Ordinal
   D. Nominal
2. If groups of participants are selected to represent sub-group in the population eg. Such as selecting an entire class of nursing to be compared to a group of health records and information class, this is known as :-
   A. Cluster sampling
   B. Multi-stage sampling
   C. Quota sampling
   D. Simple random sampling
3. The number of tally sheet count for each value or a group is called?
   A. Class limit
   B. Class width
   C. Frequency
   D. Class boundary
4. Classification of data by attributes is called?
   A. Quantitative classification
   B. Qualitative classification
   C. Chronological classification
   D. Geographical classification
5. In a given set of data, if you divide the class frequency by the total number of observations you get
   A. Relative frequency
   B. Cumulative frequency
   C. Bivariate frequency
   D. Percentage frequency
6. In a relative frequency distribution, the sum of the relative frequency is equal to?
   A. 100
   B. One
   C. ΣF
   D. ΣX
7. Presentation of data in form of rows and columns is called?
   A. Array presentation
   B. Tabulation
   C. Stem and leaf plot
   D. Data analysis
8. Histogram can be drawn only for:
   A. Relative frequency distribution
   B. Cumulative frequency distribution
   C. Discrete frequency distribution
   D. Continuous frequency distribution
9. Cumulative frequency polygon cab be used for calculation of;
   A. Mean
   B. Median
   C. Mode
   D. correlation
10. Which measure of central tendency is greatly affected by extreme values?
    A. Mean
    B. Median
    C. Mode
    D. Range
11. What percentage of scores falls within one standard deviation from the mean in a normal distribution
    A. 99.7%
    B. 50%
    C. 68%
    D. 95%
12. Which one of the following statements is always correct?
    A. Mode = 2 median – 3 mean
    B. Mean = median = mode
    C. Median = Q2 = D5 = P50
    D. Mean # median # mode

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13. If there are five values each equal to 16, then the standard deviation of these values is?
    A. 0
    B. 16
    C. 4
    D. 1
14. The variance of a sample of 25 observations equals 16. The standard deviation of the samples is equal to?
    A. 5
    B. 4
    C. 9
    D. 41
15. What does a large standard deviation suggest?
    A. The scores are widely distributed
    B. The scores are evenly distributed
    C. The distribution of the scores is skewed
    D. There is strong positive correlation between two variables
16. When the correlation coefficient, r, is close to negative one (-1)
    A. There is a weak linear relationship between the two variables
    B. There is a strong linear relationship between the two variables
    C. There is no linear relationship between the two variables
    D. There is a weak negative relationship between the two variables
17. Which two measures use the mean as a baseline and identify the extent to which scores differ from this
    A. Mode and median
    B. Range and interquartile range
    C. Correlation and regression
    D. Variance and standard deviation
18. _____ can be simply defined as lack of symmetry in the distributions of scores
    A. Variation
    B. Dispersion
    C. Skewness
    D. Deviation
19. The ratio of the standard deviation to the arithmetic mean expressed as a percentage is called?
    A. Coefficient variation
    B. Coefficient standard deviation
    C. Efficient skewness
    D. Kurtosis
20. The measurement of speed or scatter of the individual values around the central point is called?
    A. Measure of skewness
    B. Measure of dispersion
    C. Measure of kurtosis
    D. Measure of central tendency
21. The following are the characteristics of standard normal distribution. Which one is not?
    A. It is bell-shaped
    B. It is bimodal
    C. The curve is symmetric about the mean
    D. The mean, median and mode are equal
22. The alternative hypothesis for the chi-square test of independence is that the variables are?
    A. Related
    B. Dependent
    C. Independent
    D. Significant

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23. What is statistical inference?
    A. The process of drawing conclusions about a sample base on population data
    B. The process of drawing conclusion about a statistic based on a parameter
    C. The process of drawing conclusion about population based on a parameter
    D. The process of drawing conclusion about a population based on a sample data

24. If you have data on the monthly average temperatures of Nairobi for the year 2018, and you are particularly interested in change over time, what is the most effective graphical displays?
    A. Histogram
    B. Scatter plot
    C. Ogive
    D. Line graph

25. Which of the following set of graphical techniques can be used to present nominal data?
    A. Bar chart and histogram
    B. Pie hart and ogive
    C. Bar chart and pie chart
    D. Histogram and ogive

26. You ask ten of your classmates about their weights. On the basis if this information you conclude that the average weight of all health records and information students is 57 kg. this is an example of
    A. Descriptive statistics
    B. Statistical inference
    C. Experimental statistics
    D. Parameter

27. Consider the following variables
    i. The colour of your hair
    ii. Your body temperature
    iii. The distance o travelled to reach your class
    In the given order, these variables are

    The above variables can be respectively categorized as;
    A. Ordinal, interval, nominal
    B. Nominal, ratio, interval
    C. Nominal, interval, ratio
    D. Nominal, nominal, interval

28. The number of accidents in a given town in the year 2017 is an example of?
    A. Discrete variable
    B. Continuous variable
    C. Qualitative variable
    D. Constant

29. The number of patients who are occupying hospital beds during given period of time is referred to as?
    A. In patient
    B. In patient days
    C. Bed safe
    D. Average length of stay

30. The continuous permanent and compulsory recording of the accurate of vital events like births and deaths is referred to as?
    A. Vital statistics
    B. Civil registration
    C. Census
    D. National sample survey

31. Which one of the following tools of measurement has time dimension?
    A. Count
    B. Proportion
    C. Rate
    D. Ratio

32. The number of live births per 1000 population in a given year is the;
    A. General fertility rate
    B. Crude birth rate
    C. Total fertility rate
    D. Crude death rate

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33. Which one of the following does not lead to the increase of prevalence rate?
    A. Increase in incidence
    B. Prolongation of life with treatment
    C. Emigration of new cases
    D. Longer duration of disease

34. Which one of the following is the correct formula for Karl Pearson's coefficient of Skewness (Skp)
    A. $SKp = \frac{\text{mean - mode}}{\sigma}$
    B. $SKp = \frac{\text{mean + mode}}{\sigma}$
    C. $SKp = \frac{\text{mean + mode}}{\text{median}}$
    D. $SKp = \frac{\text{mode + mean + median}}{3}$

35. Graphical methods may be used to determine the value of the following measures except
    A. Median
    B. Mode
    C. Correlation
    D. Mean

36. Data arranged in ascending or descending order of magnitude is called?
    A. Arrayed data
    B. Ungrouped data
    C. Grouped data
    D. Discrete frequency distribution

37. In as statistical table, the heading of the rows are called?
    A. Titles
    B. Captions
    C. Stubs
    D. Body

38. To predict the value of one variable (the dependent variable) on the basis of other variables (independent variables) we use
    A. Measures of variation
    B. Correlation coefficient
    C. Regression analysis
    D. Coefficient of skewness

39. A parameter is a measure which is computed from
    A. Primary data
    B. Secondary data
    C. Sample data
    D. Population data

40. Questionnaire survey method is used to collect
    A. Secondary data
    B. Primary data
    C. Qualitative variable
    D. Quantitative variable

PART II: SHORT ANSWER QUESTIONS (40 MARKS)

41. A student sat for exams in five paper and score the following, 64, 56, 70, 67, 73. Calculate the variance and standard deviation of the scores (5 marks)

42. In the year 2018, Mwangaza hospital which had 45 beds allocated throughout the year recorded 11,680 in-patient days. A total of 98 patients died while 2,238 were discharged during the periods. From the above information, calculate,
    a) Available bed days (1 mark)
    b) % occupancy (1 mark)
    c) Average daily vacant beds(1 mark)
    d) ALOS (1 mark)
    e) TOB (1 mark)

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43. Health statistics is the science of collecting, organizing, presenting and interpretation of health data. List the sources of data for health statistics. (5 marks)

44. Indicate the numerators and denominators used while calculation the following vital statistics measures (5 marks)

S/No	Measure	Numerator	Denominator
a.	Disease prevalence rate		
b.	Child-woman ratio		
c.	Infant mortality rate		
d.	Crude death rate		
e.	Age-specific birth rate		

45. Define the following term as used in health statistics (5 marks)
   a) Census
   b) Gravidity
   c) Fecundity
   d) Variable
   e) Live birth

46. State five uses of health statistics (5 marks)

47. Calculate the geometric mean for the following data 10, 12, 12, 13, 15, 19 (5 marks)

48. List five vital events that ate included in civil registration system (5 marks)

PART III: LONG ANSWER QUESTIONS (20 MARKS)

49. The following table shows the data recorded from the heights of trees measured in centimeters. Use the information to answer the questions below

Length of trees (cm)	frequency
150 -154	5
155 – 159	2
160 – 164	6
165 – 169	8
170 – 174	10
175 – 179	11
180 – 184	5
185 - 189	3

   a) Calculate the mean height of the trees (5 marks)
   b) Calculate the median height of the trees (5 marks)
   c) Estimate the mode of the trees by use of a suitable graphical method (10 marks)

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